CSC343 Assignment 3 – Part 2 Solutions

1. Relation $R_1(L, M, N, O, P, Q, R, S)$

Given FDs:

$$S_1 = \{L \to NQ,\ MNR \to O,\ O \to M,\ NQ \to LS,\ S \to OPR\}$$

(a) BCNF violations

An FD $X \to Y$ violates BCNF iff X is not a superkey of R_1. We compute the closure of each LHS in S_1:

- $L^+ = \{L\} \xrightarrow{L \to NQ} \{L, N, Q\} \xrightarrow{NQ \to LS} \{L, N, Q, S\} \xrightarrow{S \to OPR} \{L, N, O, P, Q, R, S\} \xrightarrow{O \to M} \{L, M, N, O, P, Q, R, S\}$. L is a superkey.
- $(NQ)^+ = \{N, Q\} \xrightarrow{NQ \to LS} \{L, N, Q, S\} \xrightarrow{S \to OPR} \{L, N, O, P, Q, R, S\} \xrightarrow{O \to M} \{L, M, N, O, P, Q, R, S\}$. NQ is a superkey.
- $(MNR)^+ = \{M, N, R\} \xrightarrow{MNR \to O} \{M, N, O, R\} \xrightarrow{O \to M} \{M, N, O, R\}$. Not a superkey.
- $O^+ = \{O\} \xrightarrow{O \to M} \{M, O\}$. Not a superkey.
- $S^+ = \{S\} \xrightarrow{S \to OPR} \{O, P, R, S\} \xrightarrow{O \to M} \{M, O, P, R, S\}$. Not a superkey.

BCNF-violating FDs (LHS is not a superkey):

$$MNR \to O, \quad O \to M, \quad S \to OPR.$$

(b) BCNF decomposition

Step 1. Start with $R_1(LMNOPQRS)$. Choose violating FD $O \to M$ (where $O^+ = \{M, O\}$).

- $R_{11}(MO)$ (attributes in O^+)
- $R_{12}(LNOPQRS)$ (R_1 attributes minus M, keeping O)

Projected FDs on $R_{12}(LNOPQRS)$: $\{L \to NQ,\ NQ \to LS,\ S \to OPR\}$.

($MNR \to O$ and $O \to M$ cannot project onto R_{12} since $M \notin R_{12}$.)

Check BCNF in R_{12}: L and NQ are still superkeys (same closures as before, restricted to R_{12}). But $S^+ = \{S\} \to \{O, P, R, S\}$, so S is not a superkey. Thus $S \to OPR$ violates BCNF in R_{12}.

Step 2. Decompose R_{12} using $S \to OPR$ (where S^+ restricted to R_{12} is $\{O, P, R, S\}$):

- $R_{121}(OPRS)$
- $R_{122}(LNQS)$

Verify each final relation is in BCNF:

- $R_A(LNQS)$: $L^+ = \{L, N, Q, S\} = R_A$ and $(NQ)^+ = \{L, N, Q, S\} = R_A$. Both LHS are superkeys. BCNF.
- $R_B(MO)$: $O^+ = \{M, O\} = R_B$. Superkey. BCNF.
- $R_C(OPRS)$: $S^+ = \{O, P, R, S\} = R_C$. Superkey. BCNF.

Final BCNF decomposition (relations in alphabetical order):

$$R_A(LNQS), \quad R_B(MO), \quad R_C(OPRS).$$

Projected FDs:

- On $R_A(LNQS)$: $L \to NQ,\ NQ \to LS$.
- On $R_B(MO)$: $O \to M$.
- On $R_C(OPRS)$: $S \to OPR$.

(c) Dependency preservation

The union of all projected FDs is: $\{L \to NQ,\ NQ \to LS,\ O \to M,\ S \to OPR\}$.

We check whether $MNR \to O$ can be derived from these. Compute $(MNR)^+$ under the projected FDs:

$$\{M, N, R\} \xrightarrow{\text{no FD fires}} \{M, N, R\}.$$

M, N, and R never appear together in any single decomposed relation, so no projected FD has its LHS contained in $\{M, N, R\}$. Since $O \notin \{M, N, R\}$, the FD $MNR \to O$ is **lost**.

The decomposition is **not dependency-preserving**.

(d) Chase test for lossless join

Decomposition: $\{R_B(MO),\ R_C(OPRS),\ R_A(LNQS)\}$.

Assign a lowercase letter to each attribute (l, m, n, o, p, q, r, s for L, M, N, O, P, Q, R, S). Each row gets letters for the attributes in its relation and arbitrary numbers elsewhere.

Initial tableau:

	L	M	N	O	P	Q	R	S
MO	1	m	2	o	3	4	5	6
$OPRS$	7	8	9	o	p	10	r	s
$LNQS$	l	11	n	12	13	q	14	s

Apply $S \to OPR$: Rows $OPRS$ and $LNQS$ agree on $S = s$, so O, P, R must match. Copy from $OPRS$: in row $LNQS$, replace $12 \to o$, $13 \to p$, $14 \to r$.

	L	M	N	O	P	Q	R	S
MO	1	m	2	o	3	4	5	6
$OPRS$	7	8	9	o	p	10	r	s
$LNQS$	l	11	n	$\mathbf{o}$	$\mathbf{p}$	q	$\mathbf{r}$	s

Apply $O \to M$: All three rows agree on $O = o$, so M must match. Copy from MO: in row $OPRS$, replace $8 \to m$; in row $LNQS$, replace $11 \to m$.

	L	M	N	O	P	Q	R	S
MO	1	m	2	o	3	4	5	6
$OPRS$	7	$\mathbf{m}$	9	o	p	10	r	s
$LNQS$	l	$\mathbf{m}$	n	o	p	q	r	s

Row $LNQS$ now has a letter for every attribute: l, m, n, o, p, q, r, s. No further FD applications produce new changes.

Chase test **passed**. The decomposition is **lossless**.

2. Relation $R_2(A, B, C, D, E, F, G, H)$

Given FDs:

$$S_2 = \{AB \to C,\ C \to ABD,\ CFD \to E,\ E \to B,\ BF \to EC,\ B \to DA\}$$

(a) Minimal basis

Step 1: Split RHS into single-attribute FDs.

$$\{\ AB \to C,\ C \to A,\ C \to B,\ C \to D,\ CDF \to E,\ E \to B,\ BF \to C,\ BF \to E,\ B \to A,\ B \to D\ \}$$

Step 2: Remove extraneous LHS attributes.

Only FDs with |LHS| > 1: $AB \to C$, $CDF \to E$, $BF \to C$, $BF \to E$.

- $AB \to C$: try removing A. $B^+ = \{B\} \xrightarrow{B \to D} \{B, D\} \xrightarrow{B \to A} \{A, B, D\} \xrightarrow{AB \to C} \{A, B, C, D\}$. $C \in B^+$, so A is extraneous. Replace with $B \to C$.
- $CDF \to E$: try removing C. $(DF)^+ = \{D, F\}$. $E \notin$. Keep C.
  Try removing D. $(CF)^+ = \{C, F\} \xrightarrow{C \to B} \{B, C, F\} \xrightarrow{B \to D} \{B, C, D, F\} \xrightarrow{BF \to E} \{B, C, D, E, F\}$. $E \in (CF)^+$, so D is extraneous. Replace with $CF \to E$.
  Re-check $CF \to E$: $F^+ = \{F\}$, $E \notin$; keep C. $C^+ = \{C\} \to \{B, C\} \to \{A, B, C, D\}$, $E \notin$; keep F.
- $BF \to C$: try removing F. $B^+ = \{B\} \xrightarrow{B \to C} \{B, C\}$. $C \in B^+$, so F is extraneous. This gives duplicate $B \to C$; remove this FD.
- $BF \to E$: $F^+ = \{F\}$, $E \notin$; keep B. $B^+ = \{B\} \to \{B, C\} \to \{A, B, C, D\}$, $E \notin$; keep F.

After Step 2: $\{B \to C,\ C \to A,\ C \to B,\ C \to D,\ CF \to E,\ E \to B,\ BF \to E,\ B \to A,\ B \to D\}$

Step 3: Remove redundant FDs.

Number the FDs after Step 2: (1) $B \to C$, (2) $C \to A$, (3) $C \to B$, (4) $C \to D$, (5) $CF \to E$, (6) $E \to B$, (7) $BF \to E$, (8) $B \to A$, (9) $B \to D$.

Remove each FD one at a time; compute closure of its LHS using the *remaining* FDs. Once an FD is discarded, it stays removed.

FD	Exclude these from Step 2 when computing closure	Closure	Decision
1	1	$B^+ = \{A, B, D\}$. $C \notin B^+$.	keep
2	2	$C^+ = \{A, B, C\}$. $A \in C^+$.	**discard**
3	2, 3	$C^+ = \{C\}$. $B \notin C^+$.	keep
4	2, 4	$C^+ = \{B, C, D\}$. $D \in C^+$.	**discard**
5	2, 4, 5	$(CF)^+ = \{B, C, E, F\}$. $E \in (CF)^+$.	**discard**
6	2, 4, 5, 6	$E^+ = \{E\}$. $B \notin E^+$.	keep
7	2, 4, 5, 7	$(BF)^+ = \{A, B, C, D, F\}$. $E \notin (BF)^+$.	keep
8	2, 4, 5, 8	$B^+ = \{B, C, D\}$. $A \notin B^+$.	keep
9	2, 4, 5, 9	$B^+ = \{A, B, C\}$. $D \notin B^+$.	keep

Minimal basis (in alphabetical order):

$$\boxed{B \to A, \quad B \to C, \quad B \to D, \quad BF \to E, \quad C \to B, \quad E \to B}$$

(b) All keys for R_2

Identify attributes by their LHS/RHS appearances in the minimal basis:

Attribute	Appears on LHS	Appears on RHS	Conclusion
A	–	✓	not necessarily in any key
B	✓	✓	must check
C	✓	✓	must check
D	–	✓	not necessarily in any key
E	✓	✓	must check
F	✓	–	must be in every key
G	–	–	must be in every key
H	–	–	must be in every key

F, G, H must be in every key. $(FGH)^+ = \{F, G, H\} \neq R_2$, so we need at least one of $\{B, C, E\}$.

- $(BFGH)^+ = \{B, F, G, H\} \xrightarrow{B \to A} \{A, B, F, G, H\} \xrightarrow{B \to C} \{A, B, C, F, G, H\} \xrightarrow{B \to D} \{A, B, C, D, F, G, H\} \xrightarrow{BF \to E} \{A, B, C, D, E, F, G, H\}$. Key.
- $(CFGH)^+ = \{C, F, G, H\} \xrightarrow{C \to B} \{B, C, F, G, H\} \xrightarrow{B \to A} \{A, B, C, F, G, H\} \xrightarrow{B \to D} \{A, B, C, D, F, G, H\} \xrightarrow{BF \to E} \{A, B, C, D, E, F, G, H\}$. Key.
- $(EFGH)^+ = \{E, F, G, H\} \xrightarrow{E \to B} \{B, E, F, G, H\} \xrightarrow{B \to A} \{A, B, E, F, G, H\} \xrightarrow{B \to C} \{A, B, C, E, F, G, H\} \xrightarrow{B \to D} \{A, B, C, D, E, F, G, H\}$. Key.

A, D are not on any LHS, so $AFGH$ and $DFGH$ cannot be keys. Each of $BFGH$, $CFGH$, $EFGH$ is minimal since FGH alone is not a superkey. All candidate keys:

$$\boxed{BFGH, \quad CFGH, \quad EFGH}$$

(c) 3NF synthesis

From the minimal basis, combine FDs with the same LHS:

$$B \to ACD, \quad BF \to E, \quad C \to B, \quad E \to B.$$

Create one relation per group (LHS $\cup$ RHS):

- From $B \to ACD$: $R_1(ABCD)$, key B.
- From $BF \to E$: $R_2(BEF)$, key BF.
- From $C \to B$: $R_3(BC)$, key C.
- From $E \to B$: $R_4(BE)$, key E.

Remove subset relations:

- $R_3(BC) \subseteq R_1(ABCD)$: $\{B, C\} \subset \{A, B, C, D\}$. Remove R_3.

- $R_4(BE) \subseteq R_2(BEF)$: $\{B, E\} \subset \{B, E, F\}$. Remove R_4.

Check if any remaining relation contains a candidate key for R_2:

- $R_1(ABCD)$: does not contain F, G, H. No.
- $R_2(BEF)$: does not contain G, H. No.

Neither contains a key, so add a key relation: $R_5(BFGH)$.

Final 3NF decomposition (alphabetical):

$$\boxed{R_1(ABCD), \quad R_2(BEF), \quad R_3(BFGH)}$$

- **Dependency-preserving**: by construction, every FD in the minimal basis is contained within some relation. $B \to ACD$ in R_1; $BF \to E$ in R_2; $C \to B$ in R_1 ($C, B \in ABCD$); $E \to B$ in R_2 ($E, B \in BEF$).
- **Lossless**: $R_3(BFGH)$ contains candidate key $BFGH$.

(d) Redundancy

Yes, redundancy is possible.

Consider $R_2(BEF)$. Its projected FDs are $BF \to E$ and $E \to B$. The keys of R_2 are BF and EF (since $(EF)^+ = \{E, F\} \xrightarrow{E \to B} \{B, E, F\} = R_2$).

The FD $E \to B$ violates BCNF in R_2 because E is not a superkey ($(E)^+ = \{B, E\} \neq R_2$). Although B is a prime attribute (so 3NF is not violated), the BCNF violation means redundancy can occur.

Concrete example: if e_1 determines b_1, then tuples (b_1, e_1, f_1) and (b_1, e_1, f_2) repeat the fact "$E = e_1 \Rightarrow B = b_1$" redundantly. This also creates the risk of update anomalies.